Dig Packet Reader (data.py) – Setup and Run Guide

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Overview

Setting up a new Windows PC to run data.py takes about one hour, most of it downloads (about 6 GB). Follow the sections in order: each one depends on the one before it.

data.py is a single Python program. You drop in 6000 forms, Dig Packets, 10000-35 wall thickness forms (.xlsx) and OmegaWrap quotes, and it reads them (including handwriting, with OCR on the graphics card), compares the values, and fills a copy of the composite calculator in Excel.

Checklist:

- [ ] Computer meets the requirements (NVIDIA card, enough RAM)
- [ ] Programs installed: Python 3.12, VS Code, Excel, NVIDIA driver
- [ ] Windows settings: long paths, page file, CUDA fallback
- [ ] Python packages installed in order, with the three fixes
- [ ] GPU checks pass
- [ ] data.py and the calculator in one folder, calculator unblocked
- [ ] First run completes

1. Computer requirements

The graphics card must be NVIDIA: the OCR (PaddleOCR) only runs on an NVIDIA card and stops with an error without one.

Part	Minimum (works, slower)	Recommended	Why
Windows	Windows 10 or 11, 64-bit	Windows 11	Excel automation and the app need Windows
Graphics card	NVIDIA with 6 GB (e.g. RTX 2060)	NVIDIA with 12–16 GB	A full run peaked at 9.5 GB of graphics memory

Part	Minimum (works, slower)	Recommended	Why
RAM	32 GB	64 GB	A full run used about 39 GB including Windows (RAM + page file)
Free disk space	20 GB	40 GB	About 6 GB of packages and models, plus the page file
Internet	Needed for setup and the first run		Models download on the first run

Times measured with the same 33 files: about 15 minutes on a 16 GB RTX A5000 laptop with 127 GB RAM, about 25 minutes on a 6 GB RTX 2060 PC with 32 GB RAM.

To check your card and its memory, open PowerShell and run:

```
nvidia-smi --query-gpu=name,memory.total,driver_version --format=csv
```

If `nvidia-smi` is not found, install the NVIDIA driver first (section 2).

2. Programs to install

Install these five programs before any Python packages. All are free except Excel, which comes with Microsoft 365.

Program	Where to get it	Notes
Python 3.12 (Microsoft Store)	Python 3.12 in the Microsoft Store	Use only this one. Do not also install Python from python.org (see below).
Visual Studio Code	code.visualstudio.com	Then add the Python extension
Microsoft Excel (desktop)	office.com → Install apps	Needed for WC Distances, Clock Distances and the Group buttons. The web version does not work.
NVIDIA graphics driver	nvidia.com/drivers	Any recent driver. The CUDA libraries come with the Python packages.

Program	Where to get it	Notes
Tesseract OCR (optional)	UB Mannheim Windows installer	Only reads photos on 10000-35 forms. Install to the default folder. Without it, that one check is skipped.

Only one Python. Two Pythons on one PC means packages land in one and data.py runs in the other ("No module named torch"). To check, run `where.exe python`. Only a path with `WindowsApps` should be listed. If `C:\Python312\python.exe` (or another) shows up, uninstall it in Settings → Apps → Installed apps ("Python 3.12.x (64-bit)"), then delete any leftover `C:\Python312` folder.

Only one Office. Several Microsoft 365 entries at once (for example "Microsoft 365", "Apps for business" and "Apps for enterprise") made Excel and Word crash on open. Keep one. To clean up, remove them all with Microsoft's automatic Office uninstall, restart, and install once from office.com with the account that has the license.

After installing Excel, open it once by hand and click through any sign-in or first-start windows. The background Excel that data.py starts cannot answer those.

3. Windows settings

Three settings, done once. Restart after the first two.

Long paths (needed). The Store Python keeps packages in a very deep folder, and some Paddle and torch files go past Windows' old 260-character limit. Right-click Start → Terminal (Admin), then run:

```
New-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Control\FileSystem" -
Name LongPathsEnabled -Value 1 -PropertyType DWORD -Force
```

More on this: Microsoft – maximum path length.

Page file (needed on PCs with 32 GB RAM or less). Windows needs room to promise memory to all the processes data.py starts. Too small a page file crashed runs with "BrokenProcessPool" and "The handle is invalid".

1. Start → type "View advanced system settings" → open it.
2. Performance → Settings → Advanced tab → Virtual memory → Change.

3. Tick "Automatically manage paging file size for all drives", or choose C: → Custom size → Initial 16384, Maximum 49152 (MB) → Set.
4. OK, then restart.

NVIDIA memory fallback (recommended on cards under 12 GB). Lets the graphics card borrow regular RAM instead of stopping when its own memory is full. NVIDIA Control Panel → Manage 3D settings → Global Settings → CUDA – Sysmem Fallback Policy → Prefer Sysmem Fallback → Apply.

Optional speed-up: if the RAM shows a lower speed than its rating (Task Manager → Performance → Memory → Speed), turn on XMP / A-XMP in the BIOS.

4. Python packages (in this exact order)

Run each command on its own, in a normal (not admin) PowerShell, and let it finish before the next. The order matters: installing EasyOCR before torch brings in a torch that can't use the graphics card.

1. Update pip:

```
python -m pip install --upgrade pip
```

2. torch, the graphics-card (CUDA 12.6) build, about 2.6 GB (PyTorch install page):

```
python -m pip install torch --index-url
https://download.pytorch.org/whl/cu126
```

3. Paddle, the graphics-card build, about 1.7 GB with its NVIDIA libraries (Paddle install page):

```
python -m pip install paddlepaddle-gpu -i
https://www.paddlepaddle.org.cn/packages/stable/cu126/
```

4. Everything else data.py uses:

```
python -m pip install paddleocr easyocr transformers numpy pillow pymupdf
openpyxl pywin32 tkinterdnd2 wordfreq pgeocode us extract-msg playwright
```

5. Fix: keep only one OpenCV. EasyOCR and PaddleOCR each bring a different OpenCV, and they overwrite each other's files. Remove all of them, then install the one PaddleOCR needs (it includes everything the others have):

```
python -m pip uninstall -y opencv-python opencv-python-headless opencv-contrib-python
```

```
python -m pip install opencv-contrib-python==4.10.0.84
```

6. Fix: torchvision to match torch. EasyOCR brings a torchvision without graphics-card support:

```
python -m pip install --force-reinstall --no-deps torchvision --index-url https://download.pytorch.org/whl/cu126
```

7. Fix: give Paddle the same cuDNN version as torch. torch and Paddle each bring NVIDIA's cuDNN library, and two different versions in one program crash with "WinError 127 … cudnn_cnn64_9.dll". First see torch's version:

```
python -c "import torch; print(torch.backends.cudnn.version())"
```

It prints a number like `91002`, which means 9.10.2. Install that version for Paddle (replace 9.10.2 if yours differs):

```
python -m pip install --no-deps "nvidia-cudnn-cu12==9.10.2.*"
```

pip will warn that paddlepaddle-gpu wants 9.5.1. That is expected and fine.

"Not on PATH" warnings during installs can be ignored, since every command uses `python -m`.

5. Check that it works

Run these four checks. If all four give the expected result, the setup is done.

Check	Command	Expected result
OpenCV	`python -c "import cv2; print('OpenCV', cv2.__version__)"`	`OpenCV 4.10.0`
torch on the graphics card	`python -c "import torch, torchvision; print(torch.__version__, torchvision.__version__, torch.cuda.is_available())"`	Both versions end in `+cu126`, then `True`

Check	Command	Expected result
Paddle on the graphics card	`python -c "import torch; import paddle; paddle.utils.run_check()"`	`PaddlePaddle is installed successfully!`
torch and Paddle together	`python -c "import torch; import paddle; print(torch.cuda.is_available(), paddle.device.is_compiled_with_cuda())"`	`True True`

The Paddle checks also print `INFO: Could not find files for the given pattern(s)`, `No ccache found`, `Logging before InitGoogleLogging()` and `pir` lines. Those are normal start-up messages.

The third and fourth checks load torch first on purpose, because data.py does the same. A `WinError 127` there means step 7 of section 4 is missing or has the wrong cuDNN version.

6. Files and folder

Put two files together in one folder, for example `C:\Users\<your name>\python`:

- `data.py`
- `AE_Composite_Calculator_v2.06_072926 NO TimeWarp.xlsm` (the composite calculator)

data.py finds the calculator next to itself and finds your Downloads folder by itself, so the Windows user name doesn't matter. The Group button makes its folders in Downloads.

Unblock the calculator if it came by email, Teams or a download:

1. Open it in Excel by hand once. A yellow Protected View bar or a red "macros blocked" bar means it is blocked.
2. Close Excel, right-click the file → Properties → tick Unblock at the bottom → OK.
3. Open it once more to check it opens cleanly, then close Excel.

In VS Code, open the folder, then Ctrl+Shift+P → Python: Select Interpreter → pick the Python 3.12 whose path contains `WindowsApps` or `PythonSoftwareFoundation`.

7. Running it

1. Open PowerShell (or the VS Code terminal) in the folder with data.py.
2. Start it and save everything it prints to `run_log.txt`:

```
$env:PYTHONIOENCODING="utf-8"; python -u data.py 2>&1 | Tee-Object -FilePath run_log.txt
```

Plain `python data.py` also works, but saves no log.

3. The Documents window opens. Drag the files onto it in any order, with their names as they are. Nothing needs sorting or renaming.
4. Wait. Each document shows its progress, then "read - tabs ready". Verify, WC Distances, Clock Distances and the Group buttons stay locked until every document is read and the calculator work is done.
5. When finished, close the window with its X.

Don't press Ctrl+C in the terminal while it runs: with no text selected, Ctrl+C stops the program. To copy terminal text, select it first or right-click.

The first run downloads the reading models (EasyOCR, PaddleOCR PP-OCRv5 and distilroberta-base), about a minute on a fast connection. Later runs start faster.

If the program closes by itself, send `run_log.txt` and `crash_log.txt` (both in the data.py folder) to whoever maintains data.py.

8. Troubleshooting

Every problem below happened during a real setup. Find the message you see in the first column.

What you see	Cause	Fix
`Access is denied` when pip installs	A second, all-users Python (e.g. `C:\Python312`) is being used	Remove the extra Python (section 2) and use the Store one
`Ignoring invalid distribution ~ip` (or `~p`, `~=p`)	Leftovers from an install that was cut off	Belong to the extra Python; removing it clears them

What you see	Cause	Fix
`No module named torch` while it is installed	Two Pythons: packages in one, data.py running in the other	`where.exe python` should list only WindowsApps; set the same one in VS Code
`WinError 127 … cudnn_cnn64_9.dll`	torch and Paddle carry different cuDNN versions	Section 4, step 7
`operator torchvision::nms does not exist`	torchvision without graphics-card support	Section 4, step 6
Odd OpenCV errors or a missing cv2 function	Several OpenCV packages overwriting each other	Section 4, step 5
`BrokenProcessPool` / `The handle is invalid`	Windows ran out of memory to promise (commit)	Page file (section 3)
`KeyboardInterrupt` and everything stops	Ctrl+C pressed in the terminal	Don't press Ctrl+C; close the window instead
Excel or Word crash on open; "this feature appears to be broken"	Several Microsoft 365 products installed at once	Keep one Office (section 2)
`0x800A03EC` in crash_log.txt	Excel refused to open the calculator	Unblock it and open it by hand once (section 6)
`0x80010001` in crash_log.txt	Excel was busy and asked data.py to wait	Normal; data.py waits and tries again
`INFO: Could not find files…` or a red `NativeCommandError` block	PowerShell showing a normal Paddle message in red	Harmless
`out of memory` or `CUDA error`	The graphics card ran out of memory	Turn on the NVIDIA memory fallback (section 3) and close games or VR apps

9. Keeping it working

The versions below worked together on 21–22 Oct 2026. Updating one of them can break the others, so update only on purpose and re-run the section 5 checks afterwards.

Package	Working version
Python	3.12 (Microsoft Store)
torch / torchvision	2.14.1+cu126 / 0.29.1+cu126
paddlepaddle-gpu	3.3.1
nvidia-cudnn-cu12	9.10.2.21 (matched to torch)
paddleocr	3.7.0
easyocr	1.7.2
transformers	5.20.2
opencv-contrib-python	4.10.0.84
numpy	2.3.5
pymupdf	2.0

Rules to keep:

- After reinstalling or upgrading paddlepaddle-gpu, redo section 4 step 7: pip puts cuDNN back to Paddle's own version.
- After upgrading torch, check its cuDNN version again and match Paddle's to it (section 4 step 7).
- Never install `opencv-python` or `opencv-python-headless`; keep only `opencv-contrib-python`.
- To copy a working PC's exact versions to a new one: on the working PC run `python -m pip freeze > packages.txt`, and on the new PC run `python -m pip install -r packages.txt`.